

Temporal PID of EEG Sleep

Short-window pass · 1-s windows · broadband · Bitbrain ds005555 sub-1 · 6 PSG channels · 3.5 h

Method

Continuous EEG signals (Bitbrain ds005555, sub-1, six PSG channels F3, F4, C3, C4, O1, O2, 256 Hz) from an overnight polysomnography recording are analysed for temporal information structure across **second-scale lags** ($1 \text{ s} \leq \tau \leq 30 \text{ s}$).

Preprocessing

The signal is divided into **non-overlapping windows** of duration $\Delta t = 1 \text{ s}$. Each window is independently discretised into $N_b = 4$ levels via **quantile binning** (equal-probability bins), producing an amplitude-invariant symbolic sequence. Within-window quantile binning removes any DC offset or slow drift on the window timescale — information-theoretic measures then capture temporal *pattern* rather than amplitude covariation.

Temporal Triplet Construction

For a target window at time t , two source windows are selected at lags τ_1 and τ_2 **seconds** into the past ($\tau_1 < \tau_2$, both in 1-s steps, max 30 s). The three aligned discrete vectors form a **temporal triplet** (s_1, s_2, target). Sample-by-sample alignment yields $L = \Delta t \times f_s = 256$ co-occurring symbols per triplet.

PID Decomposition

The empirical joint distribution $p(s_1, s_2, \text{target})$ over $N_b^3 = 64$ states is estimated by counting co-occurrences. **Partial Information Decomposition** (PID, using the Minimum Mutual Information measure) then decomposes the total predictive information $I(\{s_1, s_2\}; \text{target})$ into four non-negative atoms. At this pass the MMI atoms are computed in **closed form** directly from the joint counts (~500–1500× faster per call than dit; verified to 1e-15 agreement):

$$\begin{aligned} R &= \min(I(S_1; T), I(S_2; T)) \\ U_i &= I(S_i; T) - R \\ S &= I(\{S_1, S_2\}; T) - I(S_1; T) - I(S_2; T) + R \end{aligned}$$

Atom	Meaning
Redundancy	Information about the target that both past timescales provide independently
Synergy	Information available only when both past timescales are considered jointly
Unique ₁	Information exclusively carried by source 1 (shorter lag)
Unique ₂	Information exclusively carried by source 2 (longer lag)

Stage Filtering

Sleep stage is determined per window from scored annotations. Because lags reach 30 s and hypnograms are scored in 30-s blocks, a triplet can in principle span a stage boundary. At this pass the stricter CONTINUOUS_STAGE_FILTER is enabled: **every window in the span from source₁ to target must share the same scored stage**. This guarantees a triplet sits entirely inside one 30-s scored bout.

Because stages have different typical bout lengths (e.g., N3 bouts can exceed many minutes while N1 rarely exceeds 3 min), not all lag pairs are available for all stages. This creates a **coverage bias**: averaging across all available lag pairs would inflate values for short-bout stages (which only contribute high-information short lags). To correct this, all cross-stage comparisons (Plots 4, 9, 14, 16) restrict to the **common lag range** — the intersection of lag pairs present in every stage — so that per-window means are computed over identical timescale subsets.

AR(1) Baseline — stage-conditional

A sample-level AR(1) process is fit **per sleep stage**: the coefficient ϕ and noise σ are estimated on that stage's discretised windows alone. A synthetic AR(1) PID matrix is computed per stage and broadcast to each target window based on its stage label. Excess = Actual – AR(1) is then a within-stage measure of non-linear / higher-order temporal structure — isolated from across-stage spectral differences. (The prior 30-s pass used a single global AR(1); the stage-conditional baseline underwrites the new Plot 7b.)

Block-permutation null

100 permutations of the 1-s window order, with PID recomputed on the shuffled stream. At $\Delta t = 1$ s the block shuffle destroys nearly all linear autocorrelation, so atoms surviving the shuffle reflect structure on the sub-second / within-window scale. This is a direct response to feedback that 30-s block shuffling — preserving essentially all the structure of interest — was uninformative.

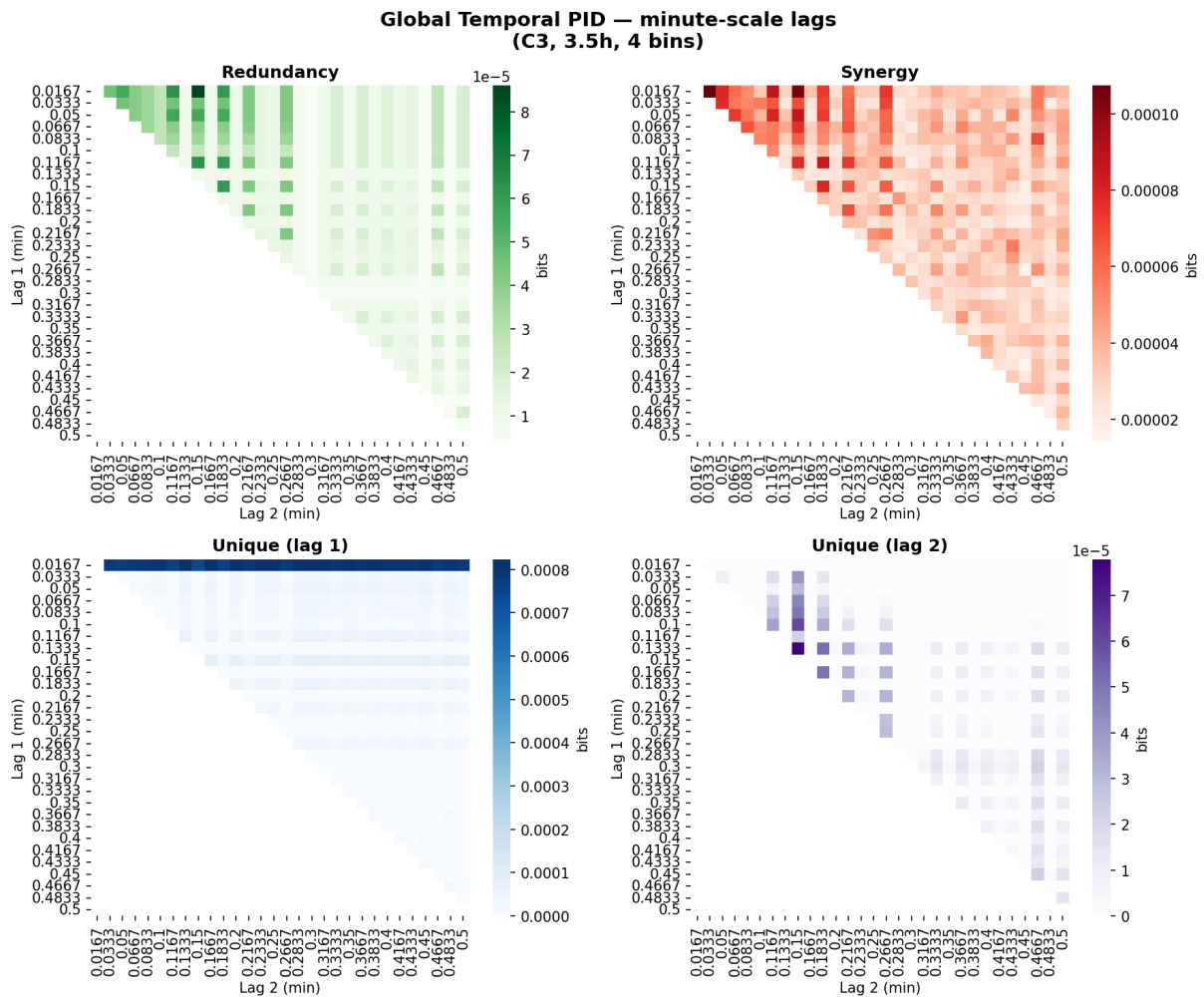
Figures

Each figure is produced per channel. The canonical instance shown below is **C3**. The same plot exists for F3, F4, C4, O1, O2 at results/pid/eeg_sleep/<channel>/<slug>_ba79f4ef.png.

Plot 1 — Global PID Matrix

File: C3/global_pid_matrix_ba79f4ef.png

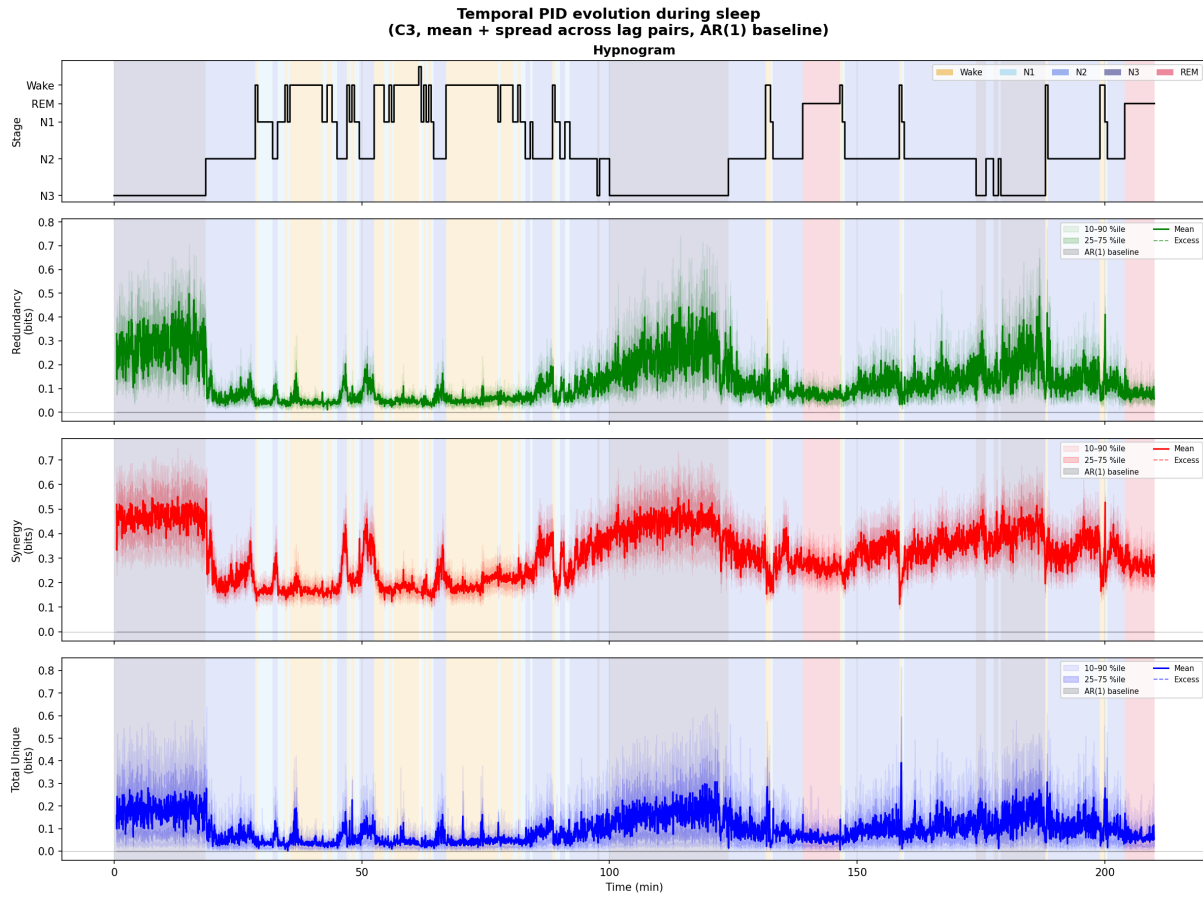
Four lag $\blacksquare$ $\times$ lag $\blacksquare$ triangular heatmaps (redundancy, synergy, unique $\blacksquare$, unique $\blacksquare$) computed over the entire recording. Each cell shows the PID atom value for that pair of lags. Reveals the overall timescale fingerprint of temporal information — which lag combinations carry redundant vs. synergistic information about the present.



Plot 2 — Hypnogram + PID Time Series

File: C3/hypnogram_pid_timeseries_ba79f4ef.png

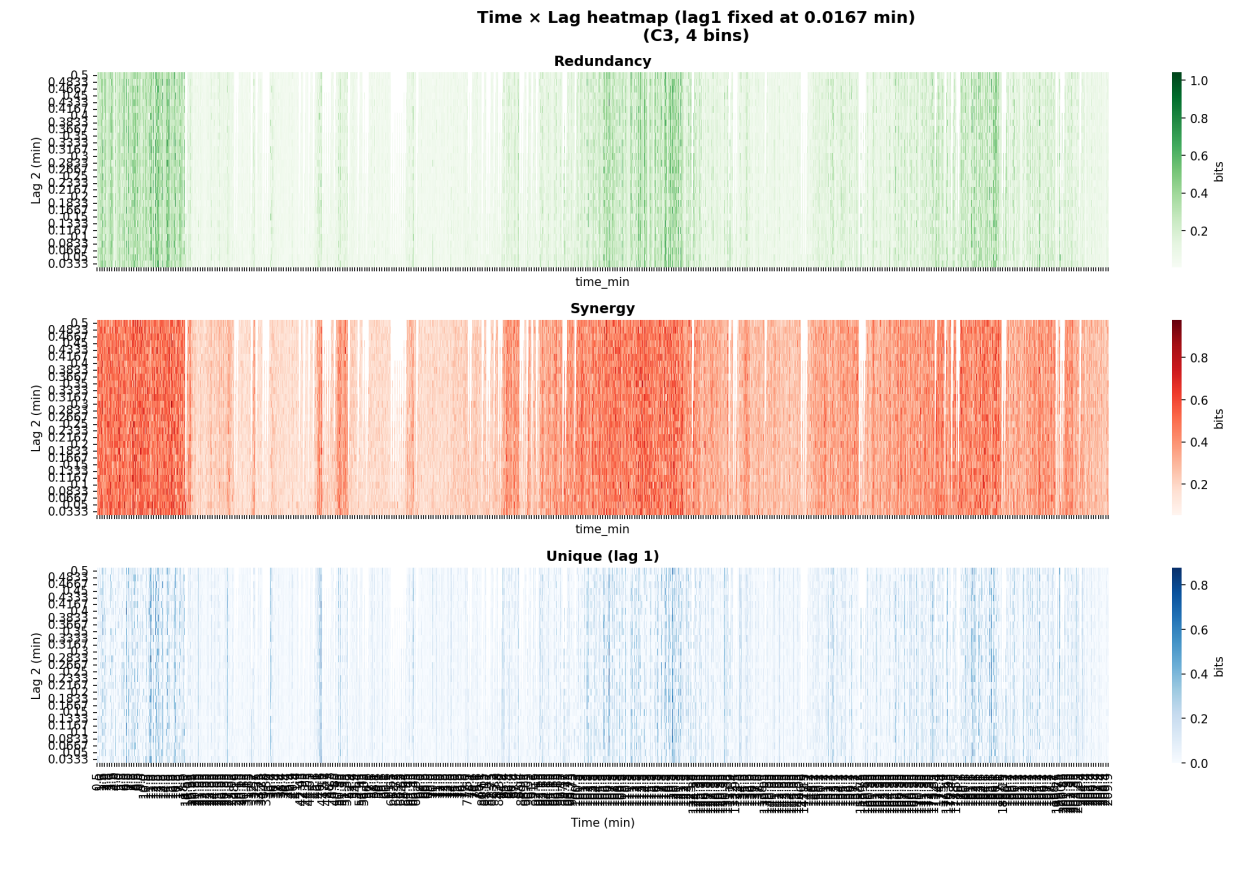
Four-panel figure. Top: hypnogram (sleep stage over time). Below: mean redundancy, synergy, and total unique information per window (averaged across all lag pairs), plotted as time series with sleep-stage background shading. Shows how temporal information structure evolves across the night and tracks sleep architecture.



Plot 3 — Time × Lag Heatmaps

File: C3/time_lag_heatmaps_ba79f4ef.png

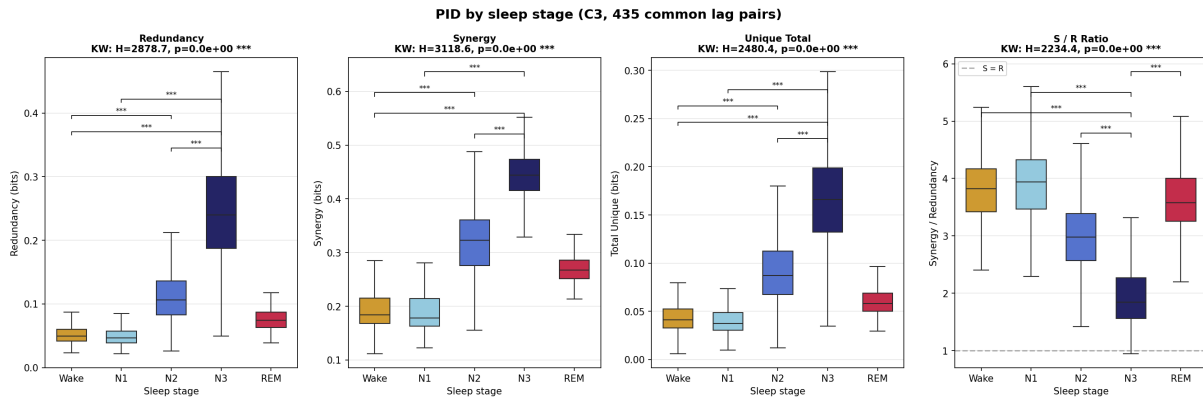
Three heatmaps (redundancy, synergy, unique) with x = time, y = lag (lag fixed at the shortest available lag). A sleep-stage colour bar is drawn above. Reveals how the timescale profile of each atom evolves over the recording — for example, whether deep sleep shows synergy at different lag scales than REM.



Plot 4 — Stage Comparison with Statistics

File: C3/stage_comparison_ba79f4ef.png

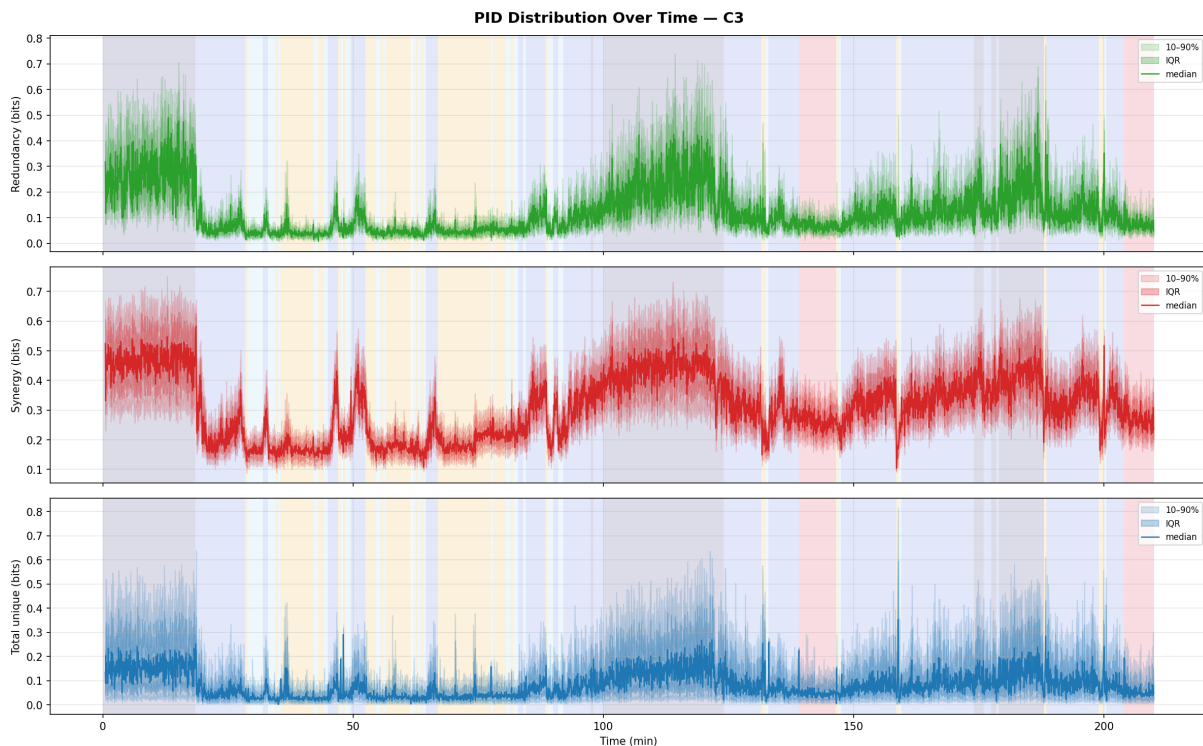
Boxplots of per-window mean PID atoms grouped by sleep stage: redundancy, synergy, total unique, and synergy/redundancy ratio. **Restricted to common lag pairs** shared by all stages to avoid coverage bias. Each panel includes a Kruskal-Wallis test (H-statistic and p-value in title) and pairwise Mann-Whitney U significance brackets for the top 4 most significant pairs. This is the primary plot for answering whether PID structure differs across sleep stages.



Plot 5 — PID Distribution Over Time

File: C3/pid_distribution_over_time_ba79f4ef.png

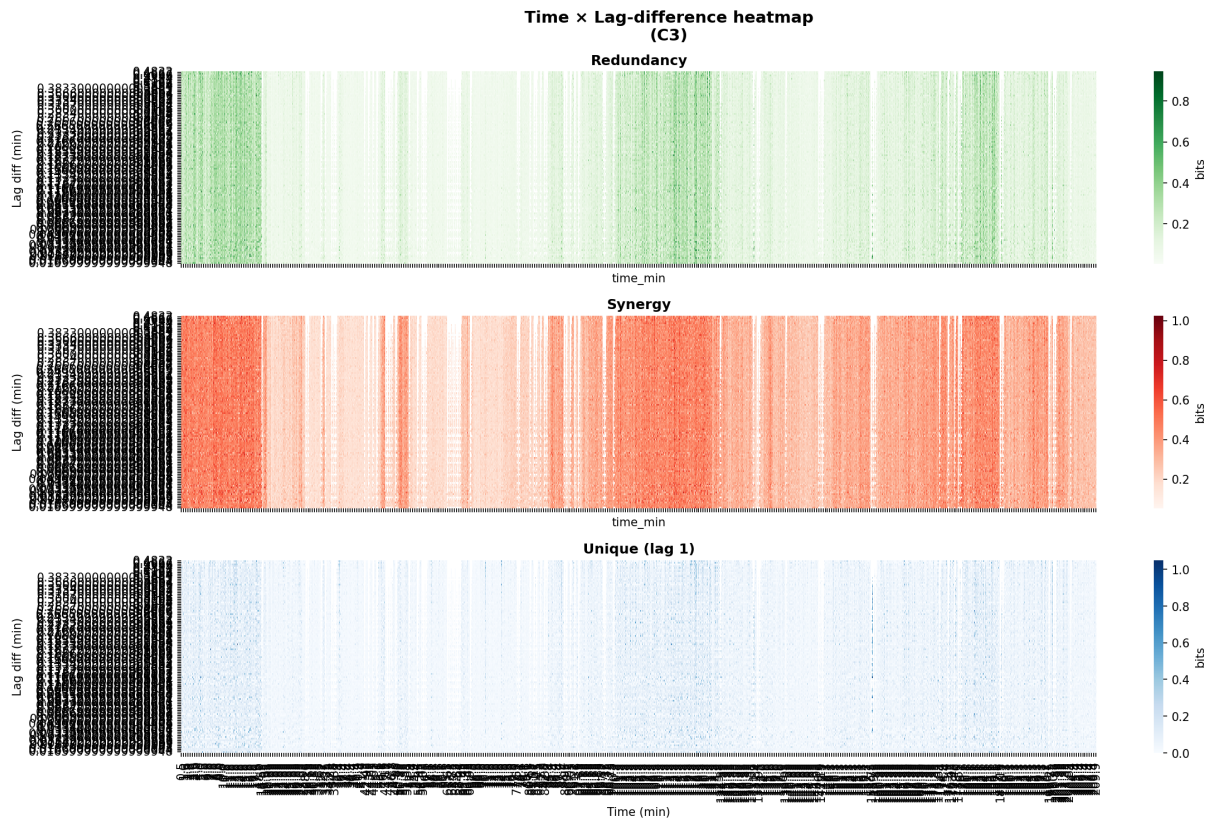
Percentile bands for each PID atom over time: for each window, the spread across all lag pairs is shown as median + IQR + 10th–90th percentile. Sleep-stage background shading is overlaid. Shows not just the central tendency but the *variability* of temporal information across timescales within each window.



Plot 6 — Lag-Difference Heatmap

File: C3/lagdiff_heatmap_ba79f4ef.png

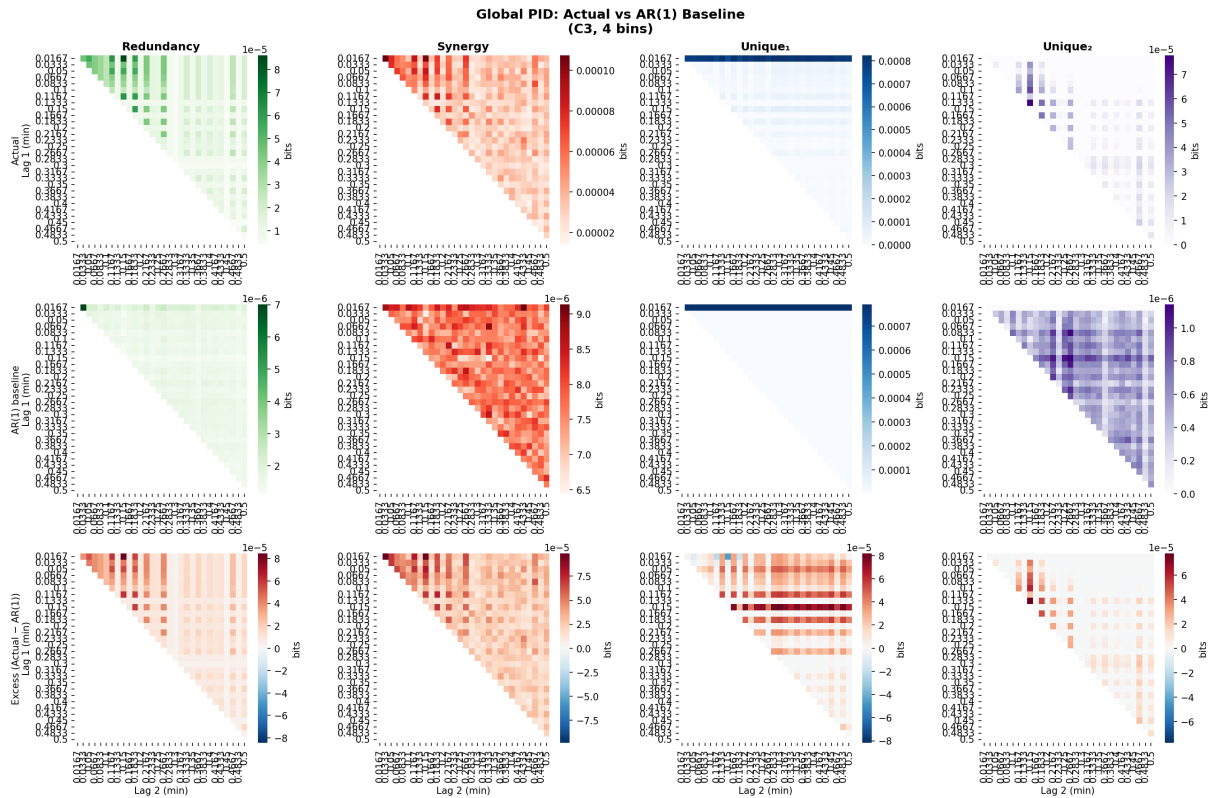
Heatmap with x = time, y = lag difference ($\text{lag} \blacksquare - \text{lag} \blacksquare$), colour = metric. Each cell is the mean metric across all lag pairs with that separation. Collapses the two-dimensional lag space into one dimension (timescale gap), making it easier to see whether certain timescale separations carry more information at specific times.



Plot 7 — Global PID: Actual vs AR(1)

File: C3/global_pid_vs_ar1_ba79f4ef.png

3 × 4 grid of lag■ × lag■ heatmaps: top row = actual PID, middle row = AR(1) baseline (averaged across the per-stage fits), bottom row = excess (actual – AR(1)). Each column is one atom. The excess row uses a diverging colourmap centred at zero. Shows whether the observed PID structure exceeds what a linear-Gaussian baseline matched to the data's marginal autocorrelation would produce. For stage-resolved excess, see Plot 7b.

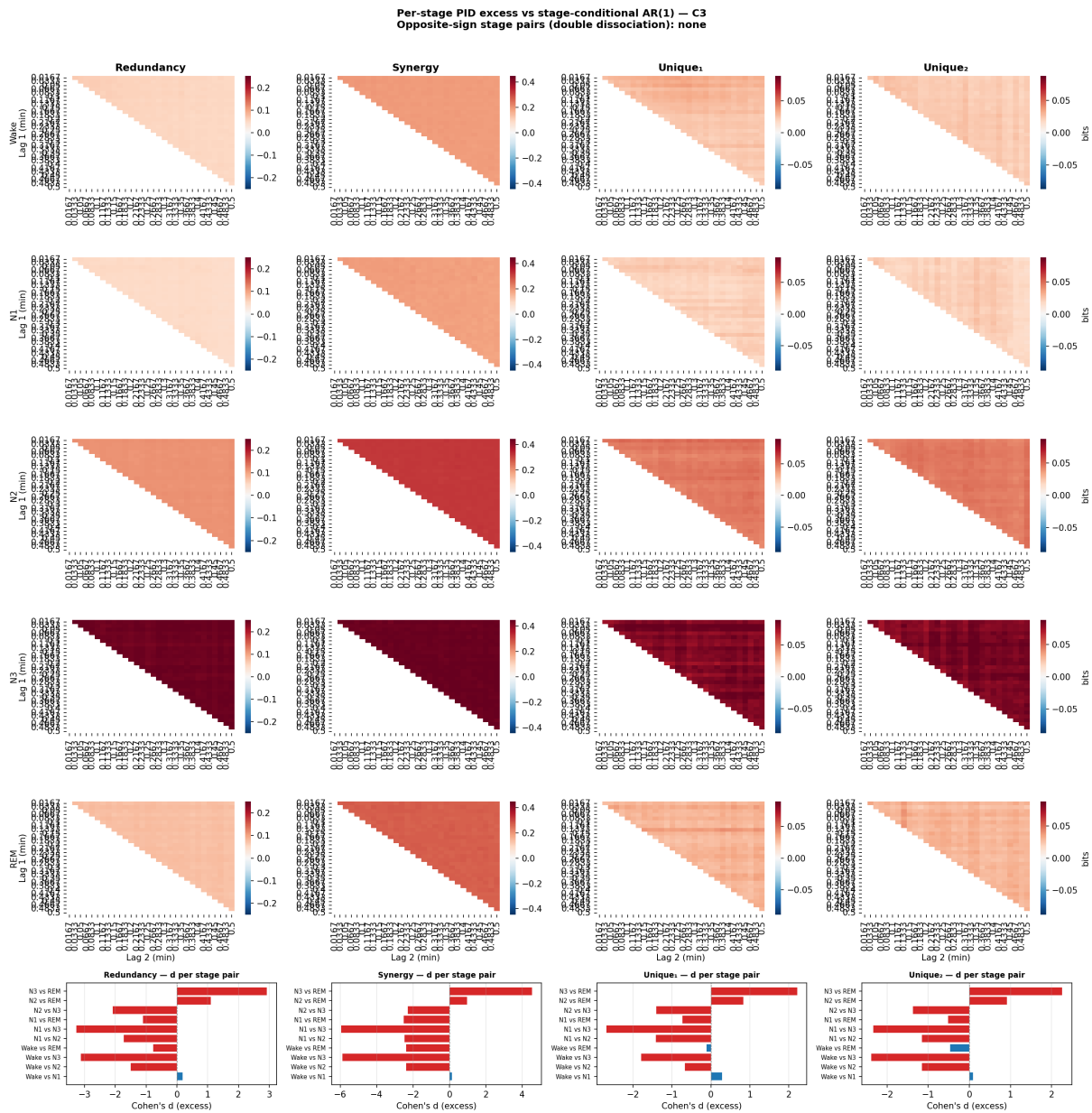


Plot 7b — Per-stage PID Excess vs Stage-Conditional AR(1)

File: C3/global_pid_vs_ar1_per_stage_ba79f4ef.png

Grid with one row per sleep stage and four columns (redundancy, synergy, unique₁, unique₂). Each cell is the lag₁ × lag₂ excess matrix (Actual – stage-matched AR(1)) for that stage and that atom, with a diverging colourmap centred at zero and a per-atom shared vmax across stages so colours are comparable.

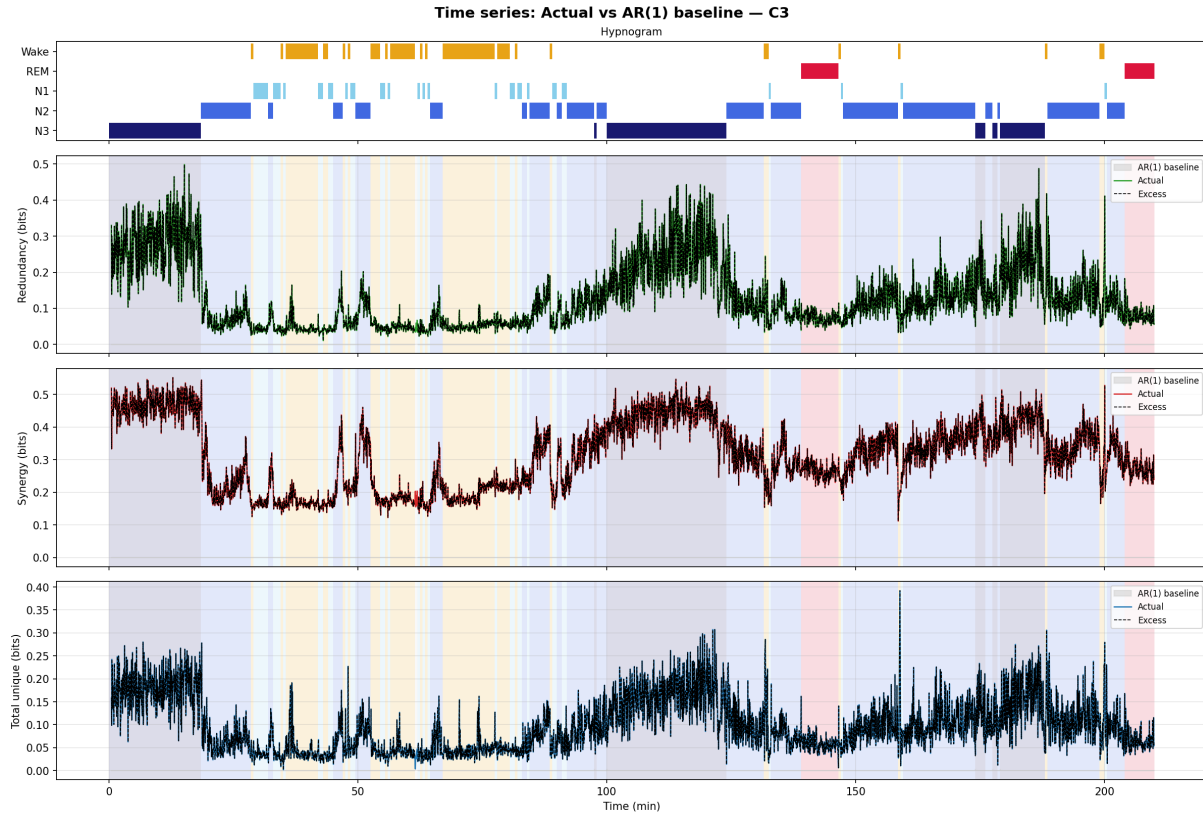
A final row carries the **double-dissociation strip**: one bar plot per atom showing Cohen's *d* of the per-window excess for every stage pair. Stage pairs whose *d* vector has opposite signs across atoms (e.g. *d* > 0 for redundancy but *d* < 0 for synergy) are highlighted in red and listed in the supitle. This is the quantitative version of the “double dissociation” Pedro flagged in Fig 7 of the prior report — it answers *where* in stage space the dissociation lives and which atom contrast carries it.



Plot 8 — Time Series vs AR(1)

File: C3/timeseries_vs_ar1_ba79f4ef.png

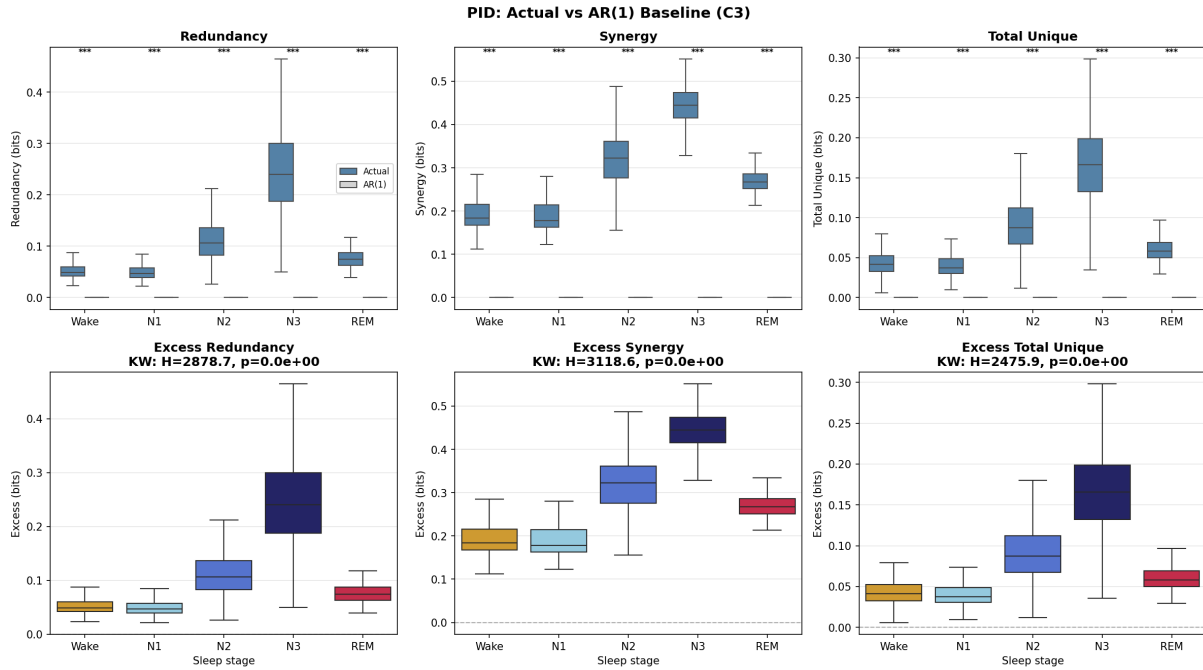
Hypnogram on top, then three panels (redundancy, synergy, total unique) showing the actual PID time series, the AR(1) baseline (grey fill), and the excess (dashed). Sleep-stage shading is overlaid. Shows the temporal evolution of non-linear temporal structure across the night.



Plot 9 — Stage Comparison: Actual vs AR(1)

File: C3/stage_comparison_vs_ar1_ba79f4ef.png

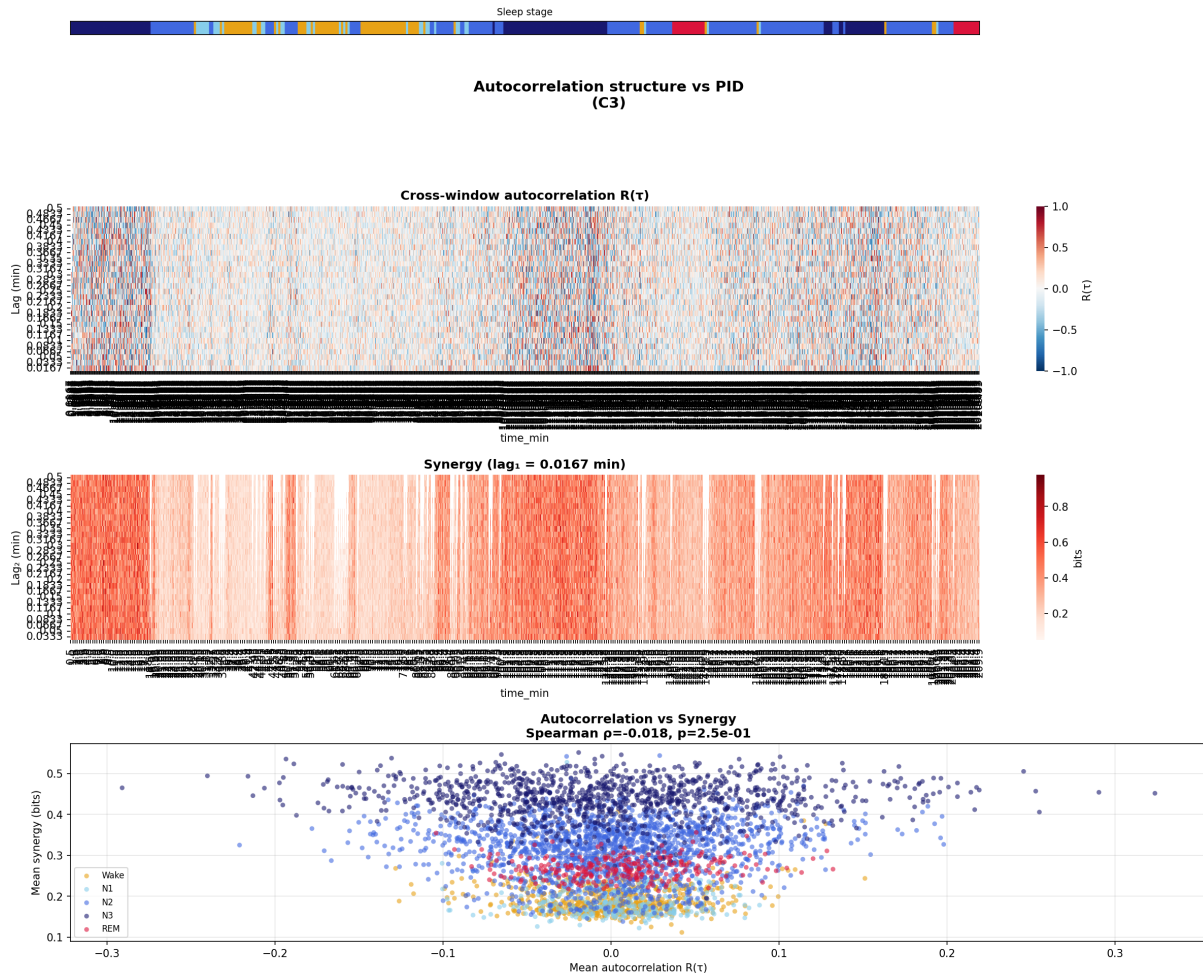
Top row: side-by-side boxplots per stage comparing actual (filled) vs AR(1) (grey) for redundancy, synergy, and total unique. Bottom row: excess (actual – AR(1)) per stage. **Restricted to common lag pairs.** Shows whether non-linear temporal structure beyond linear prediction differs across sleep stages.



Plot 10 — Autocorrelation vs PID

File: C3/autocorrelation_vs_pid_ba79f4ef.png

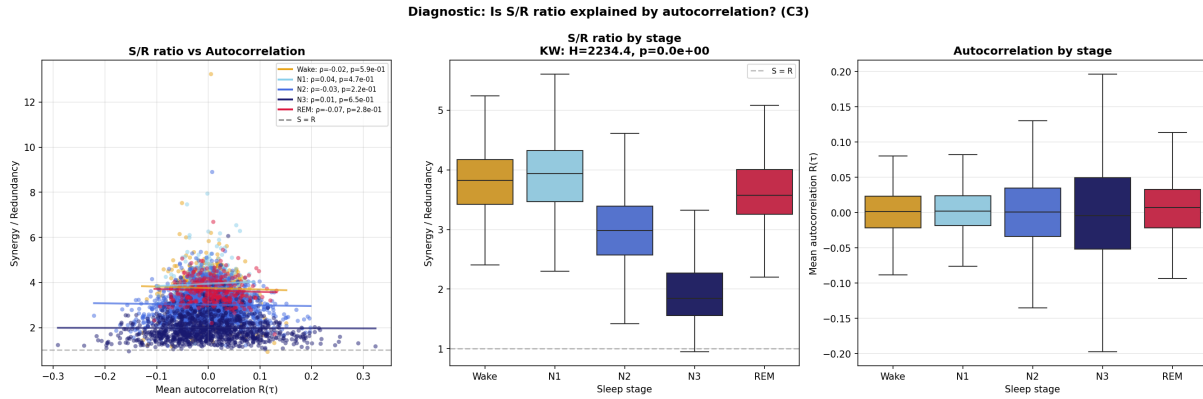
Three-panel figure. Top: cross-window autocorrelation $R(\tau)$ heatmap (time $\times$ lag). Middle: synergy heatmap at the shortest available τ (same axes for comparison). Bottom: scatter of mean autocorrelation vs. mean synergy per window, coloured by stage. Reveals whether PID changes merely track linear autocorrelation or capture something beyond it.



Plot 11 — S/R Ratio vs Autocorrelation Diagnostic

File: C3/sr_ratio_vs_autocorr_ba79f4ef.png

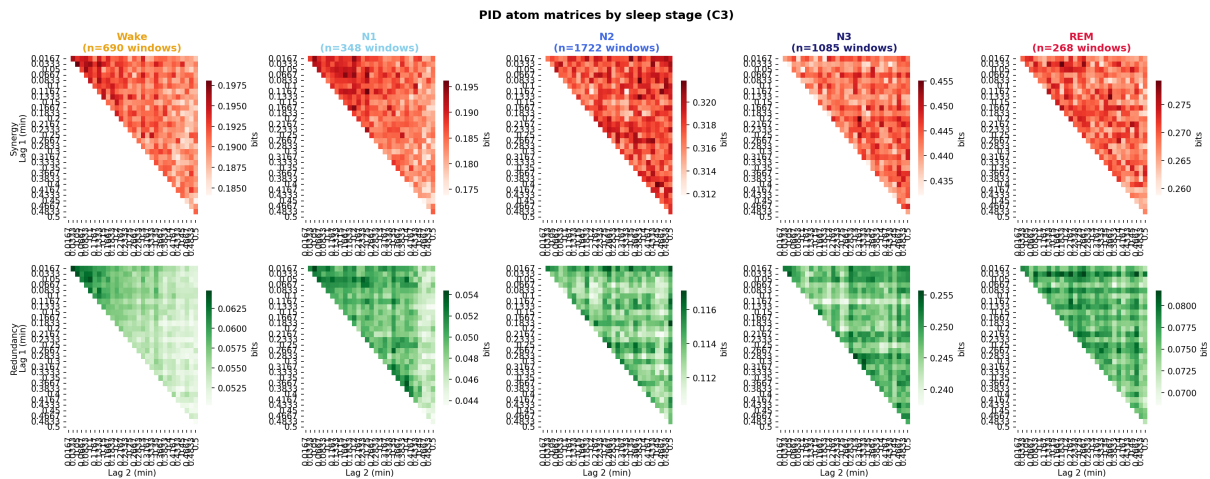
Key diagnostic figure. Three panels: (1) scatter of synergy/redundancy ratio vs mean autocorrelation, coloured by stage with regression lines; (2) S/R ratio boxplot per stage; (3) autocorrelation boxplot per stage. If stages show different S/R ratios at similar autocorrelation levels, the PID structure is genuinely different — not just an autocorrelation confound.



Plot 12 — PID Atom Matrices Per Stage

File: C3/pid_per_stage_matrix_ba79f4ef.png

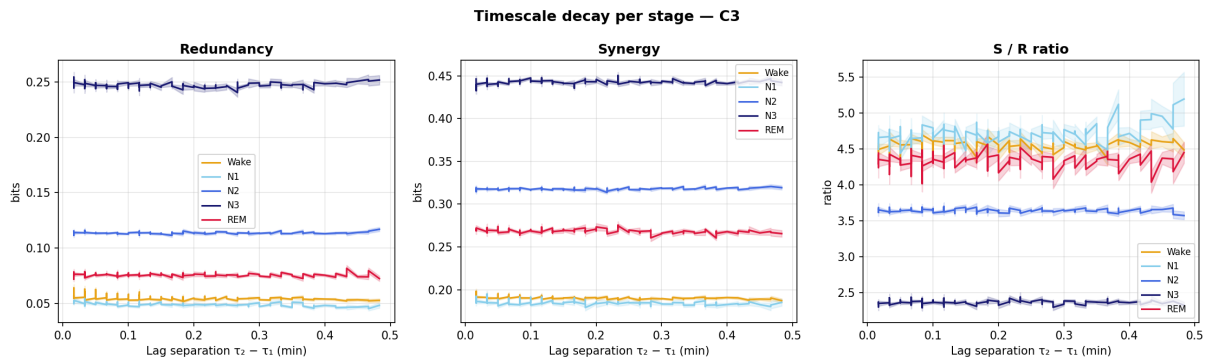
Separate $\text{lag} \times \text{lag}$ synergy and redundancy heatmaps for each sleep stage. Each column is one stage; top row = synergy, bottom row = redundancy. The number of contributing windows is shown in the title. Reveals whether stages have different *timescale fingerprints* — not just different magnitudes, but different lag-pair profiles.



Plot 13 — Timescale Decay

File: C3/timescale_decay_ba79f4ef.png

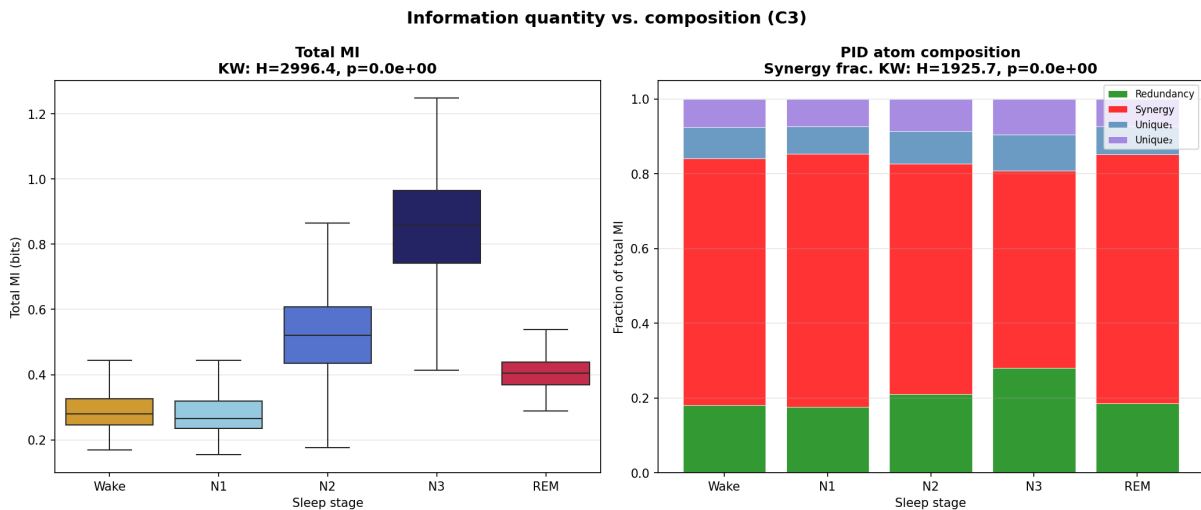
Mean PID atom as a function of lag separation ($\text{lag} \blacksquare - \text{lag} \blacksquare$), one line per stage with SEM error bands. Three panels: redundancy, synergy, and S/R ratio. Shows how fast information decays with increasing timescale gap and whether stages differ in their decay rate — e.g. does N3 (slow-wave sleep) maintain information over longer separations?



Plot 14 — Atom Fractions

File: C3/atom_fractions_ba79f4ef.png

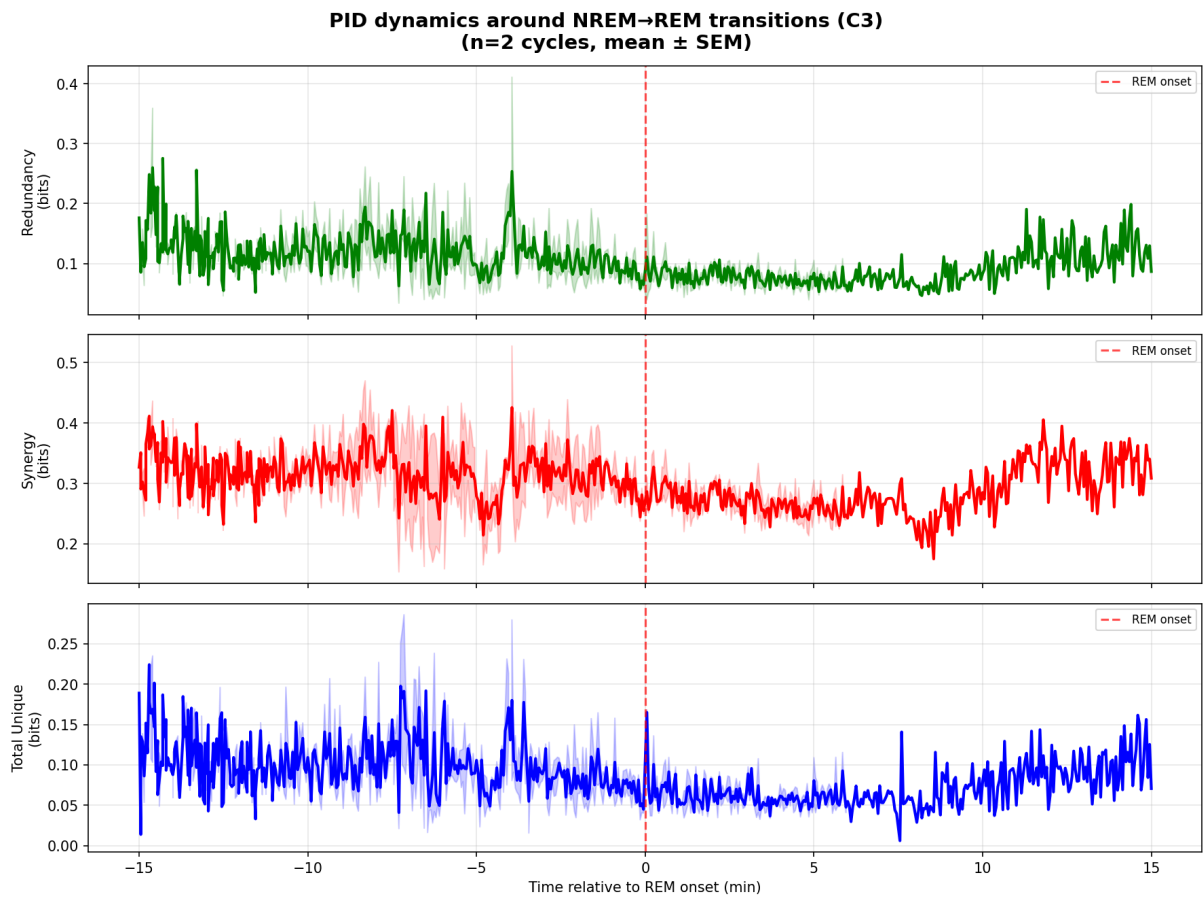
Left: total mutual information per stage (boxplot with Kruskal-Wallis statistic). Right: stacked bar chart showing the *fraction* of total MI that is redundancy, synergy, unique $\blacksquare$, and unique $\blacksquare$ per stage. **Restricted to common lag pairs**. Reveals whether stages differ in *how* they represent information (composition), not just *how much* (magnitude).



Plot 15 — NREM→REM Cycle Dynamics

File: C3/nrem_rem_cycles_ba79f4ef.png

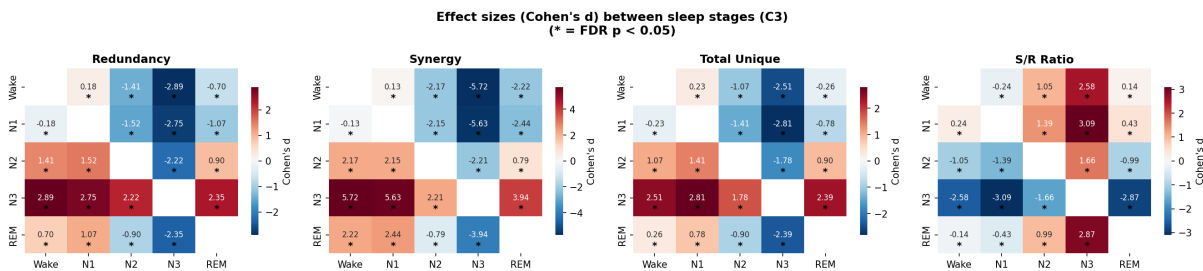
PID atoms aligned to REM onset (time = 0), averaged across all NREM→REM transitions in the recording (± 15 min window). Mean $\pm$ SEM for redundancy, synergy, and total unique. Shows whether there is a stereotyped trajectory of temporal information dynamics at sleep stage transitions — a potential signature of the NREM→REM switching process.



Plot 16 — Effect Sizes (Cohen's d)

File: C3/effect_sizes_ba79f4ef.png

Four heatmaps (redundancy, synergy, total unique, S/R ratio) showing Cohen's d between all stage pairs. Cells are annotated with the d value; asterisks mark pairs significant at $p < 0.05$ (Mann-Whitney U). Uses a diverging colourmap. **Restricted to common lag pairs.** Highlights which stage contrasts have the largest information-theoretic differences and provides standardised effect sizes for comparison.



Configuration (current pass — short windows)

Parameter	Value	Description
WINDOW_SEC	1	Window duration in seconds — addresses Pedro's point that 30-s blocks are too long for EEG
MAX_LAG_MIN	0.5	Maximum lag in minutes (= 30 s)
N_BINS	4	Number of quantile bins (reduced from 6 because each 1-s window has only 1 s · $\Delta t = 256$ samples)
DISCRETIZE_PER_WINDOW	DOWN	Per-window quantile binning
CONTINUOUS_STAGE_FILTER	FILTER	At $\Delta t = 1$ s, triplets can span a 30-s lag — requiring every interior window to share the stage
BANDS	None	Broadband first pass; band-resolved pass to follow
DURATION_HOURS	3.5	Captures ~2 full NREM–REM cycles on ds005555 sub-1 — enough N3 + ≥ 1 REM bout
TARGET_STEP	3	Stride for the time-resolved PID target axis (every 3 s). The window itself is still 1 s; only the target is stepped
Cache hash	ba79f4ef	Filename suffix on every result CSV / PNG

Stage-conditional AR(1) baseline parameters (ϕ , σ , n_{windows} per stage) are dumped per channel to `results/pid/eeg_sleep/<channel>/ar1_fit_params_ba79f4ef.json`.

Scripts

```
python scripts/pid/eeg_sleep_compute.py # all 6 channels, broadband
python scripts/pid/eeg_sleep_plot.py # all 16+1 figures × 6 channels
python scripts/utils/build_sleep_report_pdf.py # this PDF
```

Results are saved to `results/pid/eeg_sleep/<channel>/`. Each PNG/CSV carries the cache hash **ba79f4ef** as a filename suffix so this pass coexists side-by-side with the 30-s band-resolved pass without overwrite.